

The Enlarged Board in G. 1/19 also referred to T. 1265/09. The case concerned computer-implemented methods of determining an efficient schedule for a plurality of scheduled agents in a telephone call centre. The effect of an efficient schedule was found to be a business aim which did not necessarily imply any technical effects. The Enlarged Board stated that in T. 1265/09, as well as in T. 531/09, the negative assessment under Art. 56 EPC was (or could have been) based on the lack of a technical effect by the simulation rather than on the non-technical nature of the simulated system.

2.5. Schemes, rules and methods for performing mental acts, playing games or doing business

According to Art. 52(2)(c) EPC, schemes, rules and methods for performing mental acts, playing games or doing business are not to be regarded as inventions.

2.5.1 Methods for doing business

In accordance with Art. 52(2)(c) EPC, in particular methods for doing business shall not be regarded inventions within the meaning of Art. 52(1) EPC. According to Art. 52(3) EPC the patentability of such methods shall only be excluded to the extent that the application, and indeed the claimed subject-matter as this defines the matter for which protection is sought, relates to methods for doing business as such. However, where the claimed method involves technical means, it does not relate to a method for doing business as such and its patentability is therefore not excluded (see also T. 258/03, OJ 2004, 575).

Pure business methods as such are not patentable (Art. 52(2) and (3) EPC 1973). Methods only involving economic concepts and practices of doing business are not inventions within the meaning of Art. 52(1) EPC 1973. A feature of a method which concerns the use of technical means for a purely non-technical purpose and/or for processing purely non-technical information does not necessarily confer a technical character on such a method (see e.g. T. 931/95, OJ 2001, 441).

In T. 931/95 (OJ 2001, 441) the board addressed the patentability of inventions relating to business methods. Claim 1 of the main request was directed to a **method** for controlling a pension benefits program by administering at least one subscriber employer account. The board stated that if the method was technical or, in other words, had a technical character, it still might be a method for doing business, but not a method for doing business as such. However, all the features of claim 1, i.e. the individual steps defining the claimed method, were steps of processing and producing information having purely administrative, actuarial and/or financial character. Processing and producing such information were typical steps of business and economic methods. Thus the invention as claimed did not go beyond a method of doing business as such and was excluded from patentability under Art. 52(2)(c) EPC 1973 in conjunction with Art. 52(3) EPC 1973.

In T. 854/90 (OJ 1993, 669), a computer manufacturer applied to patent a method for operating an electronic self-service machine (e.g. cash dispenser) which could be accessed using any machine-readable card. A prospective user first had to insert such a card to enable its identification data to be stored; he then keyed in credit information about

himself onto an "electronic application form", thereby permitting a decision to be taken on whether to authorise him as a user. Once authorised, he could access the system using that same card. The board ruled that this was not patentable – parts of the method claimed were merely instructions for using the machine, and although technical components were used, this did not alter the fact that what was being claimed was a method for doing business as such. The board held that a claim which, when taken as a whole, is essentially a business operation, does not have a technical character and is not a claim to a patentable invention within the meaning of Art. 52(1) EPC 1973, even though the claimed method includes steps which include a technical component. The true nature of the claimed subject-matter remains the same, even though some technical means are used to perform it.

A method of doing business is excluded from patentability even where it implies the possibility of making **use of unspecified technical means** or has practical utility (T 388/04).

In T 388/04 (OJ 2007, 16) the board considered that transmitting information as specified in the claim, even if performing the transmission might involve the use of technical means, does not go beyond a business activity as such. If the contrary were true, no method for doing business that implicitly could make use of a functionality offered by technical means would be excluded from patentability, which, in the view of the board, could not be what is intended by Art. 52(2) and (3) EPC 1973. The board held that subject-matter or activities that are excluded from patentability under Art. 52(2) and (3) EPC 1973 remain so even where they imply the possibility of making use of **unspecified technical means**.

In T 384/07 the method claimed included features corresponding to technical means for the technical implementation of the business scheme, such as a data processing terminal connected to a network of data processing terminals etc. The board held that the method did not constitute a method for doing business as such, and, therefore, was not excluded from patentability in accordance with Art. 52(2) and (3) EPC 1973.

2.5.2 Methods for performing mental acts

The use of technical means for carrying out a method for performing mental acts, partly or entirely without human intervention, might, having regard to Art. 52(3) EPC 1973, render such a method a technical process or method and therefore an invention within the meaning of Art. 52(1) EPC 1973 (T 38/86, OJ 1990, 384; see also T 258/03, OJ 2004, 575).

In T 2720/16, the board held that an activity was purely mental if it was performed entirely by the human brain. As soon as it manifested itself physically outside the brain, it lost its purely mental character. It did not matter whether this physical dimension to the activity was performed using a body part (eye, hand, etc.) or separate mechanical devices.

In T 471/05 the board pointed out that claim 1 merely formulated a series of mathematical and optical abstract concepts without properly requiring a physical, technical implementation. It followed that the subject-matter for which protection is sought was the

mere "design" of an optical system and encompasses purely abstract and conceptual implementations excluded from patent protection pursuant to Art. 52(1), (2) and (3) EPC 1973. More particularly, the claimed method could be carried out as a purely mental act or as a purely mathematical design algorithm and, consequently, encompasses embodiments falling within the category of methods for performing mental acts as such and within the category of mathematical methods as such both expressly excluded from patent protection under Art. 52(2)(a) and (c) EPC 1973 in conjunction with Art. 52(3) EPC 1973.

The board stated that the criteria for technical character of a claimed invention discussed in decision T 619/02 (OJ 2007, 63) implicitly presupposed that the claimed subject-matter defining the matter for which protection is sought related to a **physical entity or a physical activity**. It could not be denied that the method defined in claim 1 of the main request could be carried out using some physical means and that such implementations of the claimed method constitute physical, technical activities not excluded from patent protection (see for instance decisions T 914/02, and T 258/03, OJ 2004, 575). Nonetheless, contrary to the appellant's contention, the claimed method **did not require the use of technical means** and the method was not restricted to physical, technical implementations, and the fact that the claimed method encompassed **non-excluded implementations** such as those mentioned above did not overcome the fact that the claimed method **also encompassed excluded subject-matter** (T 914/02, and T 388/04, OJ 2007, 16; see also T 453/91, T 930/05). Thus, as long as the claimed design method was not confined to physical, technical implementations, the claimed subject-matter encompassed embodiments excluded from patentability under Art. 52(1) to (3) EPC 1973 and was not entitled to patent protection under the EPC. The board also stated that the mere presence in the claim of such purely conceptual technical considerations did not overcome the conclusion above that the claim merely formulated a sequence of mathematical and optical concepts without properly requiring a technical or even a physical implementation. In particular, a purely mental implementation of the claimed method remains a mental act as such within the meaning of Art. 52(2) and (3) EPC 1973 even if the mental act involves conceptual technical considerations as already concluded in decision T 914/02 in which a method claim involving technical considerations and encompassing technical embodiments was refused on the grounds that the invention as claimed could still be exclusively performed by purely mental acts. In view of the above the subject-matter defined in claim 1 of the main request was excluded from patent protection under Art. 52(1), (2) and (3) EPC 1973.

In T 603/89 (OJ 1992, 230) the invention consisted of an apparatus for and a method of learning how to play a keyboard instrument, where numbers corresponding to notes on a sheet of music also appeared on the keys. The technical feature claimed was the marking of the keys. Patentability was ruled out by Art. 52(2)(c) and (d) EPC 1973. Since the key markings were merely known technical features, the contribution made by the claimed invention to the working of the teaching apparatus lay solely in the content of the information displayed, not in the apparatus itself. The invention was not based on a technical problem, but on an improvement to a teaching method, which was equivalent to an improvement to a method for performing mental acts.

The invention at issue in T. 547/14 concerned a method for predicting the formation of mould on an object, for an example a building component. The examining division had found that there was nothing to distinguish the features of claim 1, which related to an experimental determination of biological germination conditions for moulds, from a purely intellectual activity, because the experimental determination was not specified in technical terms and could be performed by purely cognitive observation. The board disagreed: if the biological germination conditions were determined experimentally, experiments had to be carried out and that meant having to use technical means; if they were determined using a computer, its use as a technical means similarly rendered the subject-matter technical. For the purpose of establishing whether the claimed method had technical character, it did not matter what details the application gave as to how the claimed experimental determination was technically implemented. That was instead a matter to be considered when assessing whether the invention could be carried out.

In T. 670/19 the claim concerned a scale system for clinical assessment of the lips and mouth area comprising a lip fullness scale comprising illustrations of the mouth area. The board held that the invention did not relate to purely abstract subject-matter and did have a technical character. The presence of a concrete physical medium, such as a board or a screen, carrying the illustrations was implied. The implicit presence of a physical medium conferred technical character to at least part of the subject-matter of the claim and this regardless of a possibly non-technical nature of the cognitive content of the illustrations carried by or displayed on the physical medium "per se".

2.5.3 Word-processing

In T. 22/85 (OJ 1990, 12) the board had to decide on the patentability of a method for automatically abstracting and storing an input document in an information storage and retrieval system and a corresponding method for retrieving a document from the system. The board observed that the described method fell within the category of activities defined in Art. 52(2)(c) EPC 1973. It considered that the mere setting out of the sequence of steps necessary to perform the activity in terms of functions or functional means to be realised with the aid of conventional computer hardware elements did not import **any technical considerations** and could therefore neither lend a technical character to the activity nor to the claimed subject-matter considered as a whole, any more than solving a mathematical equation could be regarded as a technical activity when a conventional calculation machine is used (see also T. 186/86, T. 95/86).

In T. 38/86 (OJ 1990, 384) the board first of all had to assess the patentability of a method for automatically detecting and replacing linguistic expressions which exceeded a predetermined level of understanding in a list of linguistic expressions. The board was of the opinion that a person who wished to carry out such a task using his skills and judgment would perform purely mental acts within the meaning of Art. 52(2)(c) EPC 1973; the schemes, rules and methods used in performing such mental acts were not inventions within the meaning of Art. 52(1) EPC 1973. The board stated that the use of technical means for carrying out a method, partly or entirely without human intervention, which, if performed by a human being, would require him to perform mental acts, **might**, having regard to Art. 52(3) EPC 1973, render such a method a technical process or method and

thus an invention within the meaning of Art. 52(1) EPC 1973. Since patentability was excluded only to the extent to which the patent application related to excluded subject-matter or activities as such, it appeared to be the intention of the EPC to permit patenting in those cases in which the invention involved a contribution to the art in a field not excluded from patentability. In the case in point this condition was not satisfied: once the **steps of the method for performing the mental acts** in question had been defined, the implementation of the technical means to be used in those steps involved no more than the straightforward application of conventional techniques and had therefore to be considered obvious to a person skilled in the art. If a claim for an apparatus (here, a word processing system) for carrying out a method did not specify any technical features beyond those already comprised in a claim pertaining to said method and furthermore did not define the apparatus in terms of its physical structure, but only in functional terms corresponding to the steps of that method, the board stated that the claimed apparatus did not contribute anything more to the art than the method, in spite of the fact that the claim was formulated in a different category. In such cases, if the method was excluded from patentability, so was the apparatus (see also T 71/91, T 121/85).

In T 110/90 (OJ 1994, 557) the invention was for a method of transforming a first editable document form prepared using a batch word processing system into a second editable document form for use on an interactive or batch word processing system. Following T 163/85 (OJ 1990, 379) the board found that control items (e.g. printer control items) included in a text represented in the form of digital data were characteristic of the word-processing system in which they occurred in that they were characteristic of the technical internal working of that system. Such control items therefore represented technical features of the word-processing system in which they occurred. Consequently, transforming control items which represented technical features belonging to one word-processing system into those belonging to another word-processing system constituted a method of a technical nature.

T 95/86 related to an invention that involved text editing on a computer display. The board stated that the activity of editing a text is principally concerned with linguistic and layout features of a text. The method as such aims at solving a problem which is essentially of a non-technical nature. The board, therefore, found that the activity of text editing as such must be considered as falling within the category of schemes, rules and methods for performing mental acts and therefore as excluded from patentability under Art. 52(2)(c) and (3) EPC (following T 186/86).

2.6. Presentations of information

Presentation of information in the sense of Art. 52(2)(d) EPC is understood as the conveying of information to a user. It concerns both the cognitive content of the information presented and the manner of its presentation (T 1143/06, T 1741/08). However, it does not extend to the technical means used for generating such presentations of information. Furthermore, conveying information to a user is to be distinguished from technical representations of information directed to a technical system which will process, store or transmit that information. Features of data encoding schemes, data structures and electronic communication protocols which represent functional data as opposed to